Welcome to ADPLS

Overview

Cryogenic probes offer exceptional sensitivity but often suffer from severe baseline distortions, making synthetic polymer ^{13}C NMR spectra difficult to correct using standard tools like TopSpin's abs or apbk, which can accidentally suppress broad polymer peaks. To overcome this, ADPLS combines the arPLS algorithm with user-defined hard anchor constraints via the ADMM optimization framework. If no regions are selected, it automatically acts as a standard arPLS algorithm.

Key Features

Interactive Hard Constraints: Easily specify regions that must be treated as baseline to protect broad polymer signals from being distorted.

Bruker TopSpin Integration: Seamless import and export support for Bruker 1D NMR binary data (1r).

Flexible Operation: Functions as either an automated baseline corrector or an interactive, user-guided tool depending on your workflow.

Licensing & Terms of Use

Academic Use: This software is free to use exclusively for academic and research purposes.

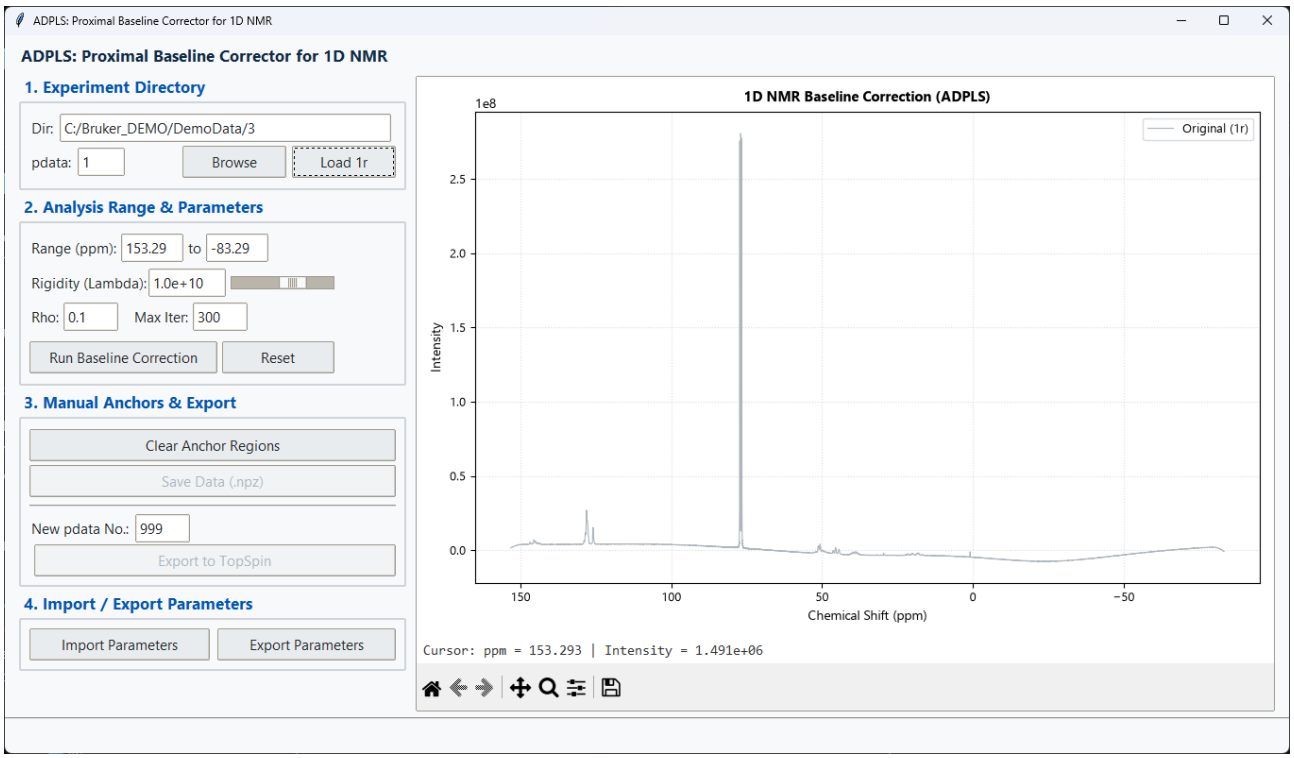
Documentation: Please check README.md and LICENSE.txt before use.

Community: Please use GitHub Discussions to exchange information, ask questions, and share usage examples with other users.

Disclaimer: This software and guide are provided "as is" for evaluation and research purposes. Please note that we cannot provide individual technical support or bug fixes.

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Loading Data in TopSpin



1. Prepare Processed Data (pdata)
Ensure your spectrum has already undergone Fourier Transformation (FT) and Phase Correction in TopSpin.

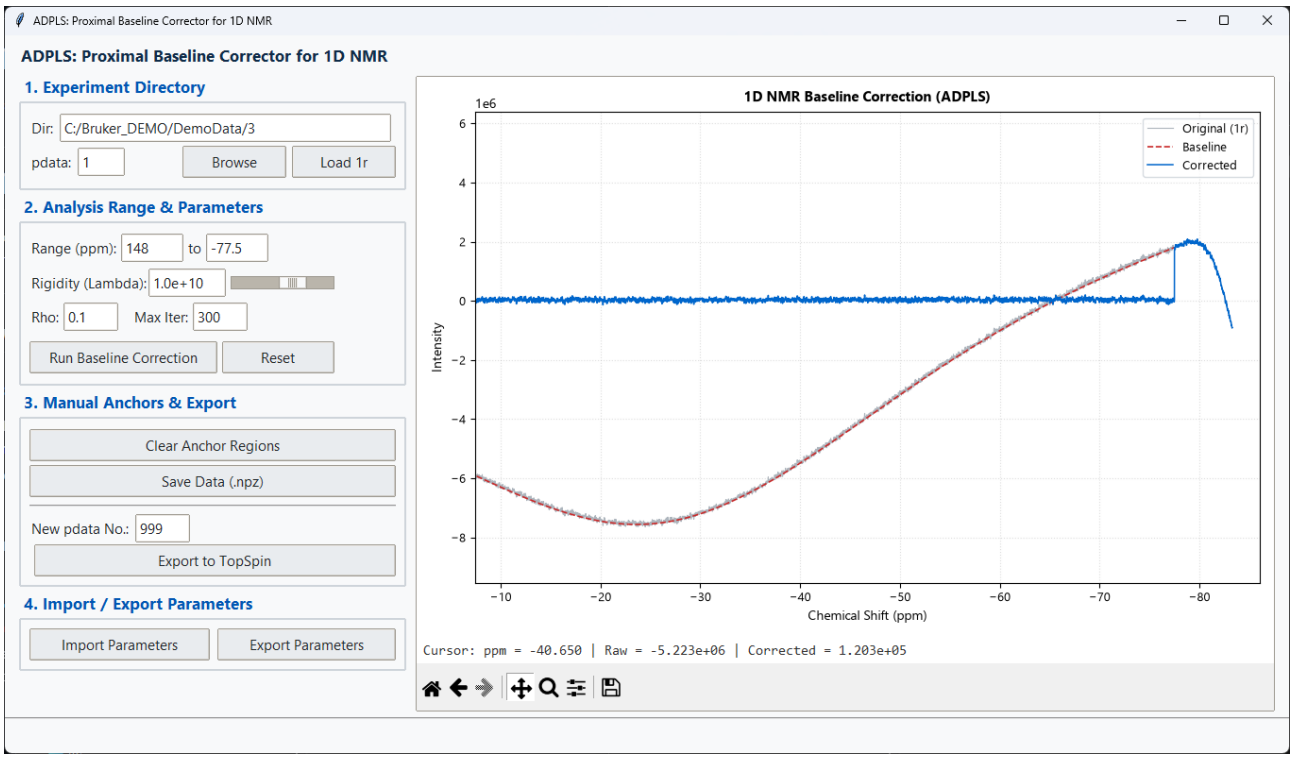
2. Set the Experiment Directory
In the Experiment Directory field, enter or browse to the path of your TopSpin experiment number directory (EXPNO).

3. Specify the pdata Number
In the pdata No. field, enter the number of the processed data directory (pdata) that contains your FT and phase-corrected spectrum (e.g., 1).

4. Load Spectrum
Click Load 1r. The loaded original spectrum will be displayed in gray on the plot area.

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Dealing with Baseline Distortion (Smiles/Frowns)



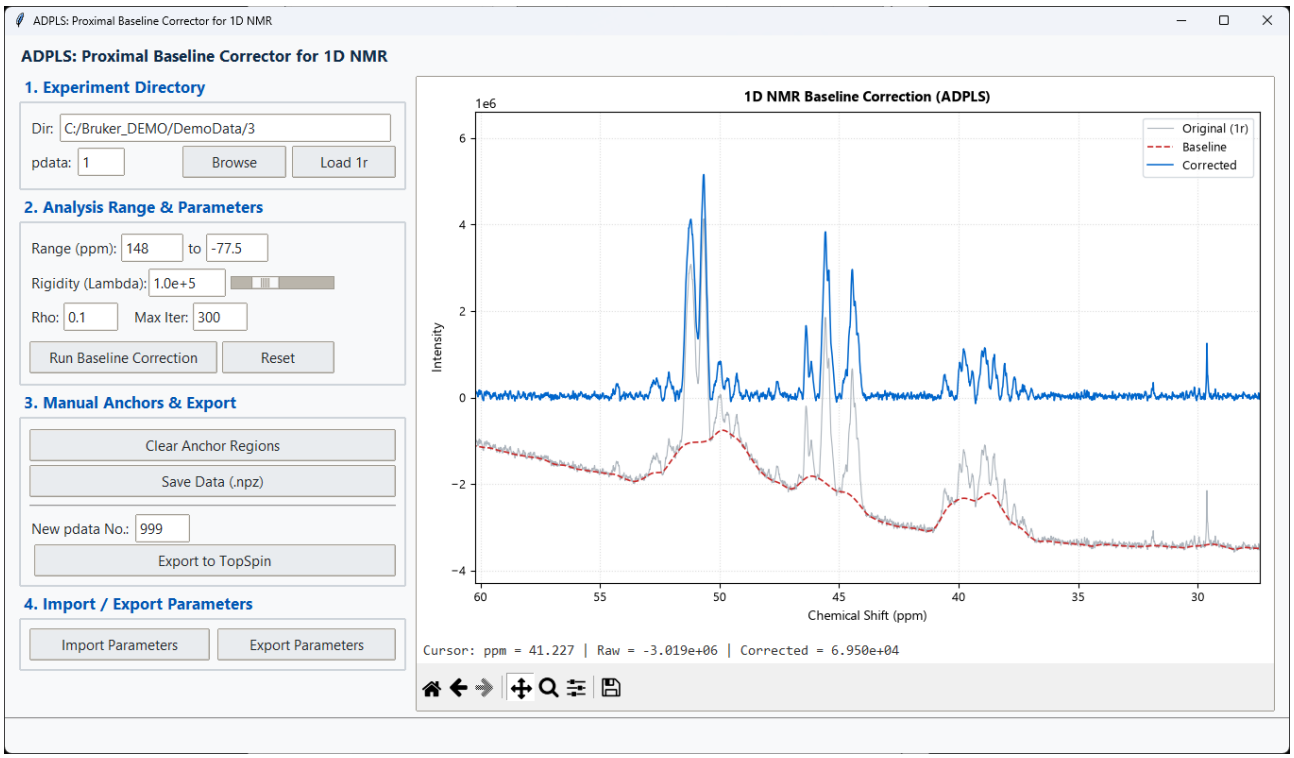
1. Understand Baseline Artifacts
Severe baseline distortions known as "smiles" or "frowns" often occur at both ends of NMR spectra.

2. Risks of Automatic Fitting
When these artifacts are fitted using the arPLS algorithm, there is a risk that even legitimate synthetic polymer peaks can be mistakenly suppressed or distorted.

3. Using Range Filtering
To mitigate this, it is very useful to specify the target baseline correction range using the Range (ppm) setting, allowing you to exclude distorted edge regions from the calculation.

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Adjusting Rigidity (Lambda)



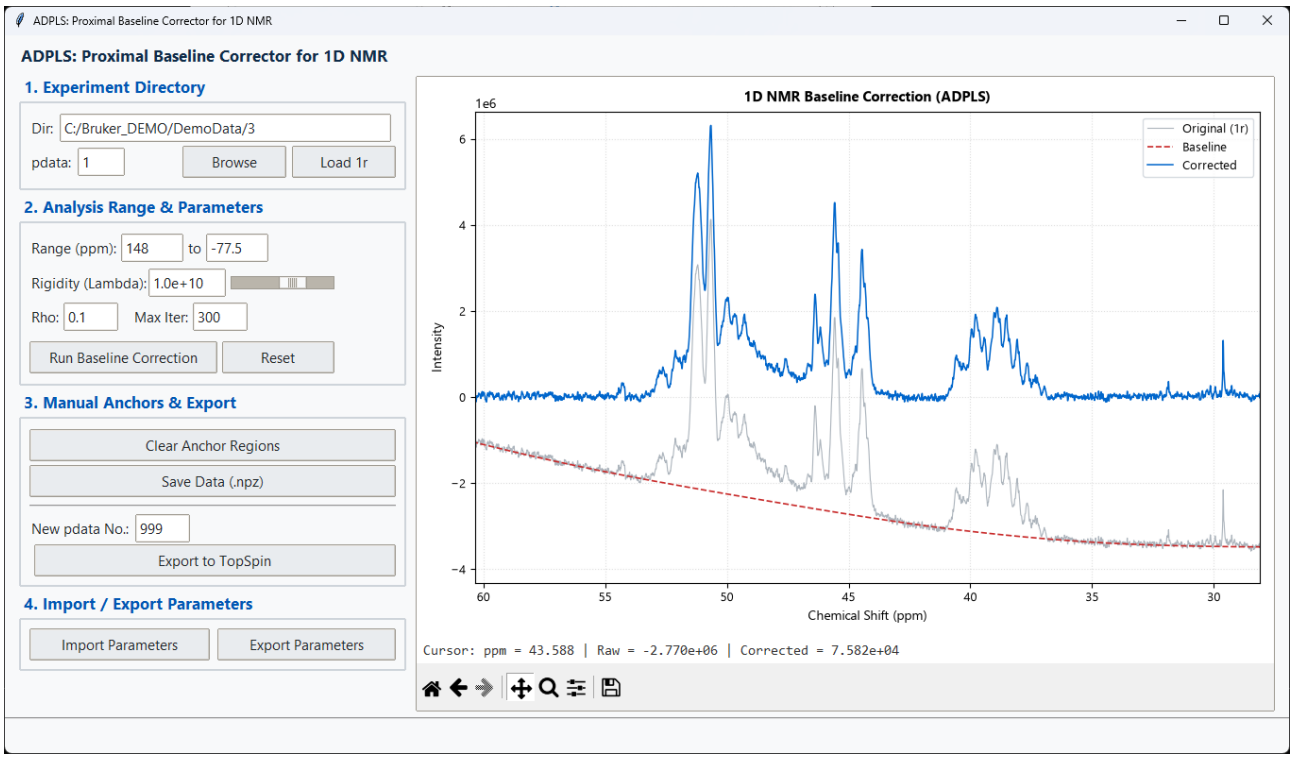
1. Adjust the Rigidity
Next, adjust the Rigidity (Lambda) parameter.

2. Understand the Effect
The smaller the Rigidity (Lambda) value, the more tightly the baseline fits to the peaks.

3. Check the Example
In the case shown in the figure above, the Rigidity (Lambda) value is too small.

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Adjusting Rigidity (Lambda) - Continued

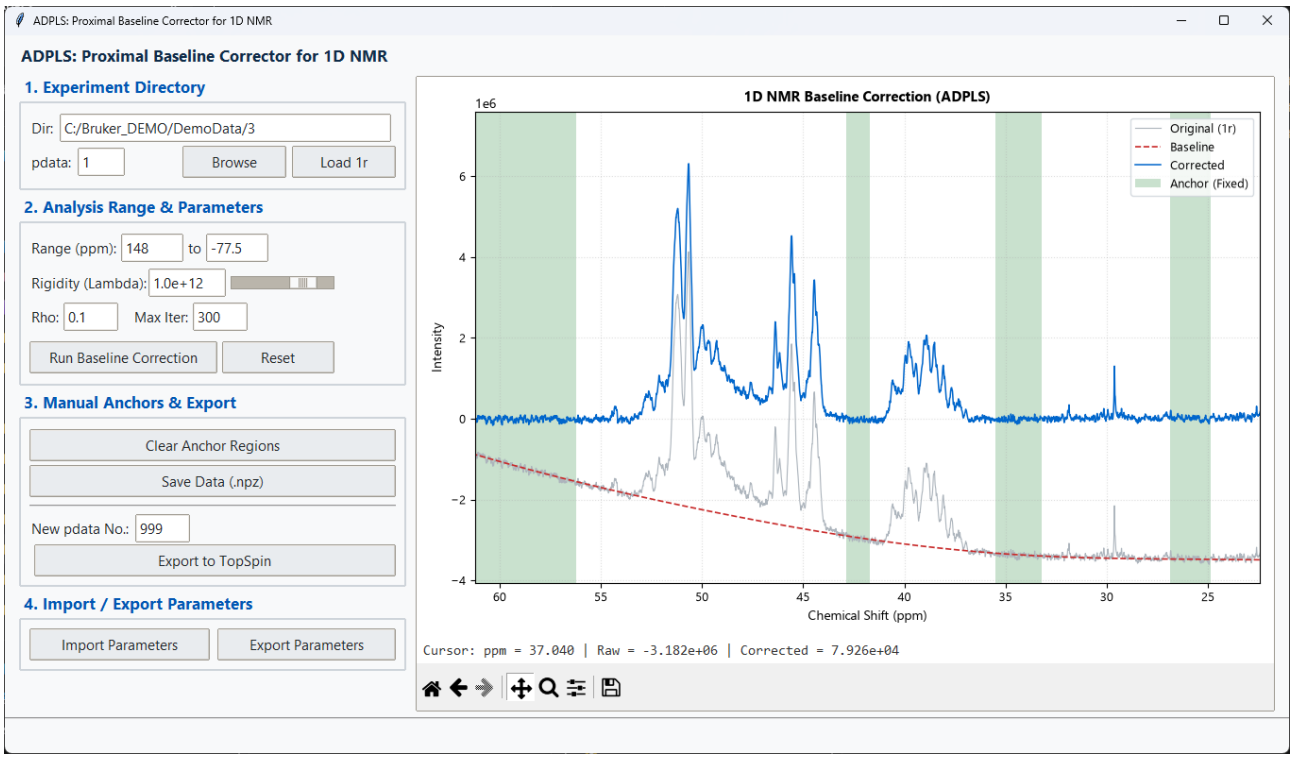


1. Increase the Value

By increasing the Rigidity (Lambda) value to an appropriate level, the algorithm successfully captures the correct baseline portion.

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Selecting Baseline Regions (Anchor Mode)



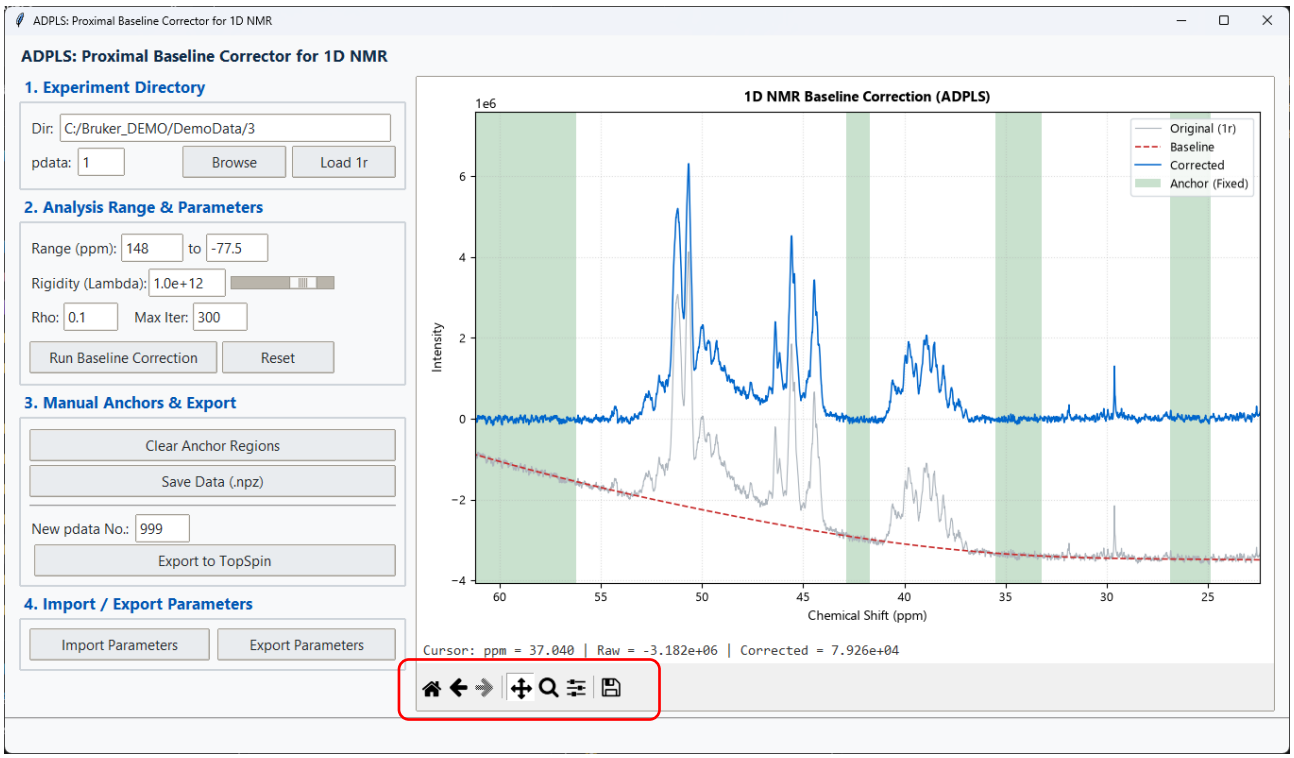
1. Specify Baseline Regions
You can optionally specify the regions you want to treat as the baseline. In the figure above, these are indicated by the green regions.

2. Add Multiple Regions
You can specify multiple regions by repeatedly left-dragging the mouse.

3. Tips for Adjustment
When using this mode, a key tip for successfully adjusting the baseline is to set a slightly larger Rigidity (Lambda) value.

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Disabling Zoom and Navigation Tools

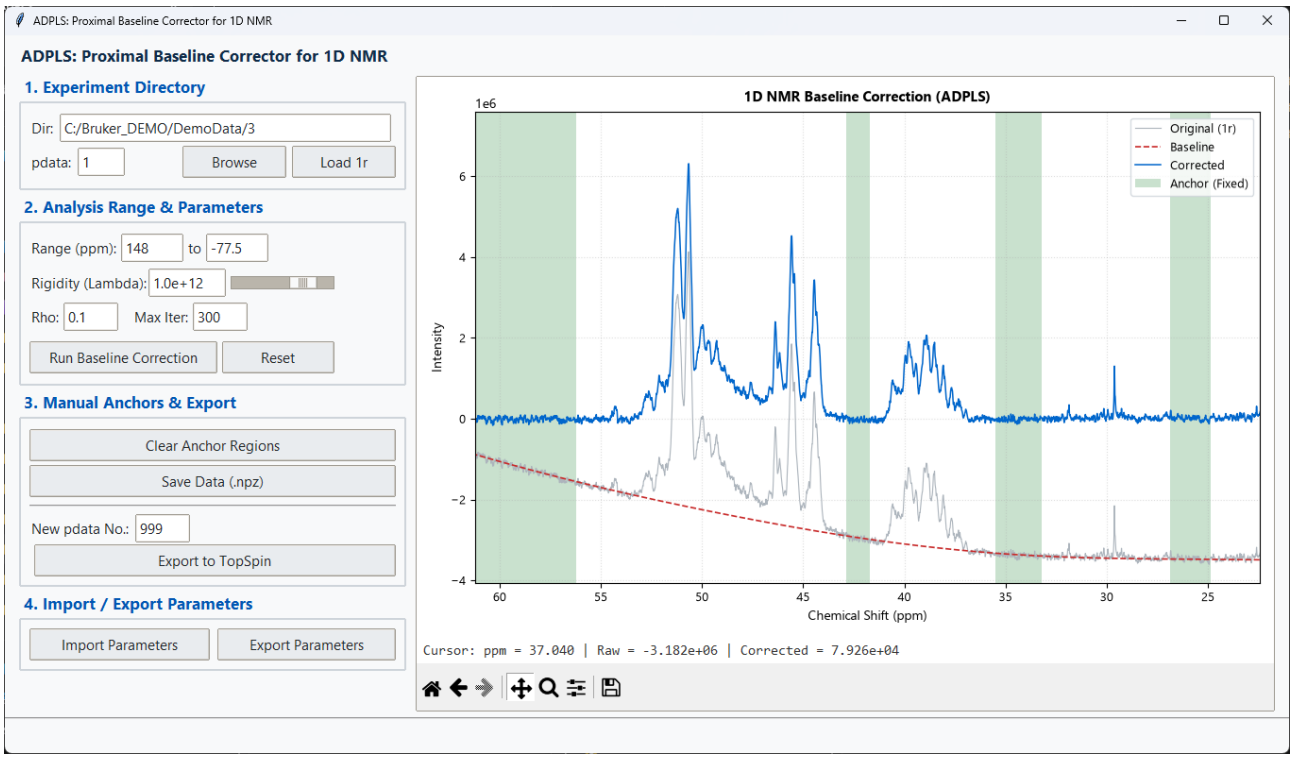


1. Check Toolbar State
If the zoom-in feature for the area enclosed by the red line is active (i.e., the button is pressed down), you cannot specify baseline regions.

2. Deactivate Tools
After zooming in, do not forget to uncheck or deactivate these toolbar buttons before attempting to select regions.

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Exporting to TopSpin



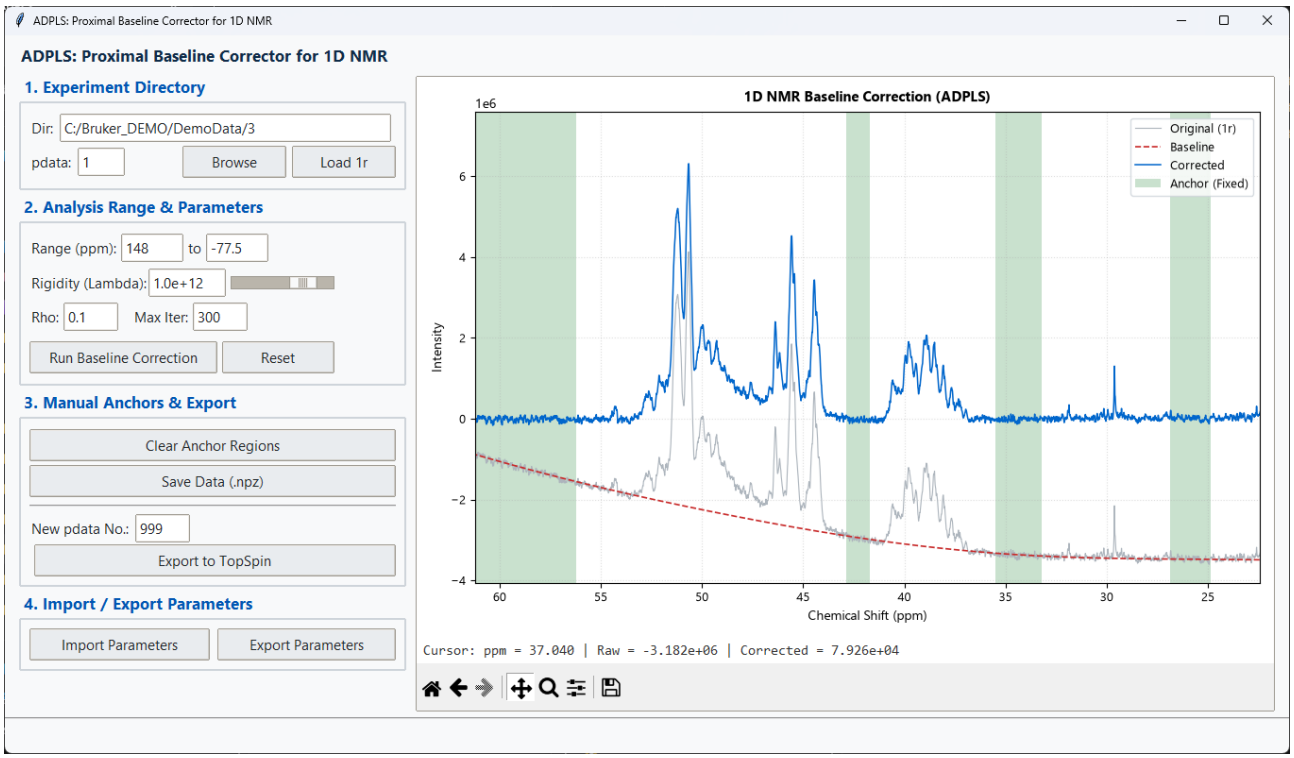
1. Click Export
Finally, click Export to TopSpin.

2. Understand the Output
In the example above, the processing conditions from pdata 1 are copied to pdata 999, and the baseline-corrected 1r file is saved into pdata 999.

3. Verify in TopSpin
Check that pdata 999 has been created in TopSpin and that you can successfully load and view the corrected spectrum.

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Advanced Options



1. Advanced Parameters
Rho (ADMM Sync) and Max Iterations are advanced options.

2. Recommended Settings
It is recommended to keep them at their default values.

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